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Surds and Indices

IMPORTANT FACTS AND FORMULAE

I. Laws of Indices:

$$(i) a^m \times a^n = a^{m+n}$$

$$(ii) \frac{a^m}{a^n} = a^{m-n}$$

$$(iii) (a^m)^n = a^{mn}$$

$$(iv) (ab)^n = a^n b^n$$

$$(v) \left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$$

$$(vi) a^0 = 1$$

II. Surds: Let a be a rational number and n be a positive integer such that $a^{\frac{1}{n}} = \sqrt[n]{a}$ is irrational. Then, $\sqrt[n]{a}$ is called a surd of order n .

III. Laws of Surds:

$$(i) \sqrt[n]{a} = a^{\frac{1}{n}}$$

$$(ii) \sqrt[n]{ab} = \sqrt[n]{a} \times \sqrt[n]{b}$$

$$(iii) \sqrt[n]{\frac{a}{b}} = \frac{\sqrt[n]{a}}{\sqrt[n]{b}}$$

$$(iv) (\sqrt[n]{a})^n = a$$

$$(v) \sqrt[m]{\sqrt[n]{a}} = \sqrt[mn]{a}$$

$$(vi) (\sqrt[n]{a})^m = \sqrt[n]{a^m}$$

SOLVED EXAMPLES

Ex. 1. Simplify : (i) $(27)^{\frac{2}{3}}$ (ii) $(1024)^{-\frac{4}{5}}$ (iii) $\left(\frac{8}{125}\right)^{-\frac{4}{3}}$.

Sol. (i) $(27)^{\frac{2}{3}} = (3^3)^{\frac{2}{3}} = 3^{(3 \times \frac{2}{3})} = 3^2 = 9$.

$$(ii) (1024)^{-\frac{4}{5}} = (4^5)^{-\frac{4}{5}} = 4^{\left\{5 \times \left(-\frac{4}{5}\right)\right\}} = 4^{-4} = \frac{1}{4^4} = \frac{1}{256}.$$

$$(iii) \left(\frac{8}{125}\right)^{-\frac{4}{3}} = \left\{\left(\frac{2}{5}\right)^3\right\}^{-\frac{4}{3}} = \left(\frac{2}{5}\right)^{\left\{3 \times \left(-\frac{4}{3}\right)\right\}} = \left(\frac{2}{5}\right)^{-4} = \left(\frac{5}{2}\right)^4 = \frac{5^4}{2^4} = \frac{625}{16}.$$

Ex. 2. What will come in place of both the question marks in the following question?

(Bank Recruitment, 2010)

$$(?)^{\frac{1}{4}} = \frac{48}{(?)^{\frac{3}{4}}}$$

Sol. Let $x^{\frac{1}{4}} = \frac{48}{x^{\frac{3}{4}}}$. Then, $x^{\frac{1}{4}} \cdot x^{\frac{3}{4}} = 48 \Leftrightarrow x^{\left(\frac{1}{4} + \frac{3}{4}\right)} = 48 \Leftrightarrow x = 48$.

Ex. 3. Evaluate : (i) $(.00032)^{\frac{3}{5}}$ (ii) $(256)^{0.16} \times (16)^{0.18}$.

Sol. (i) $(0.00032)^{\frac{3}{5}} = \left(\frac{32}{100000}\right)^{\frac{3}{5}} = \left(\frac{2^5}{10^5}\right)^{\frac{3}{5}} = \left\{\left(\frac{2}{10}\right)^5\right\}^{\frac{3}{5}} = \left(\frac{1}{5}\right)^{(5 \times \frac{3}{5})} = \left(\frac{1}{5}\right)^3 = \frac{1}{125}.$